Comparison of OLS and Ridge Regression models for House Price Prediction

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Intro

Problem statement

The main objective of this project is to create and validate a machine learning model that accurately predicts the final sale price of residential properties. This is a supervised regression model problem where the model must learn the complex relationship between a large variety of property characteristics called features and the continuous target variable of house price.The features used for the prediction contain quantitative and categorical data and the presence or absence of key amenities like pools, garages, basements, decks etc. Using theses detailed input of the model this project aims to create and train a model that generalizes and provides accurate reliable prices for new and unseen properties.

Motivation

The motivation for this project comes from the hassle and cost of the person selling their home to get a good appraisal for their house and for home buyers that want to get a good understanding of how much house they can afford. The goal of this model is to streamline the process of getting home buyers and sellers accurate prices on properties to determine how much they can sell for or how much they can afford.

Approach

This model will use the supervised learning approach to predict the continuous sales price.This is also known as a regression problem. I chose to compare the two learning algorithms Ordinary least squares and Ridge Regression. OLS will serve as the foundational linear model to establish the performance baseline to compare to the Ridge Regression which uses L2 regression will mitigate overfitting and multicollinearity among house features.The model will be evaluated using metrics such as mean square error and root mean squared error. I will compare the results to determine what is the best model.

Data

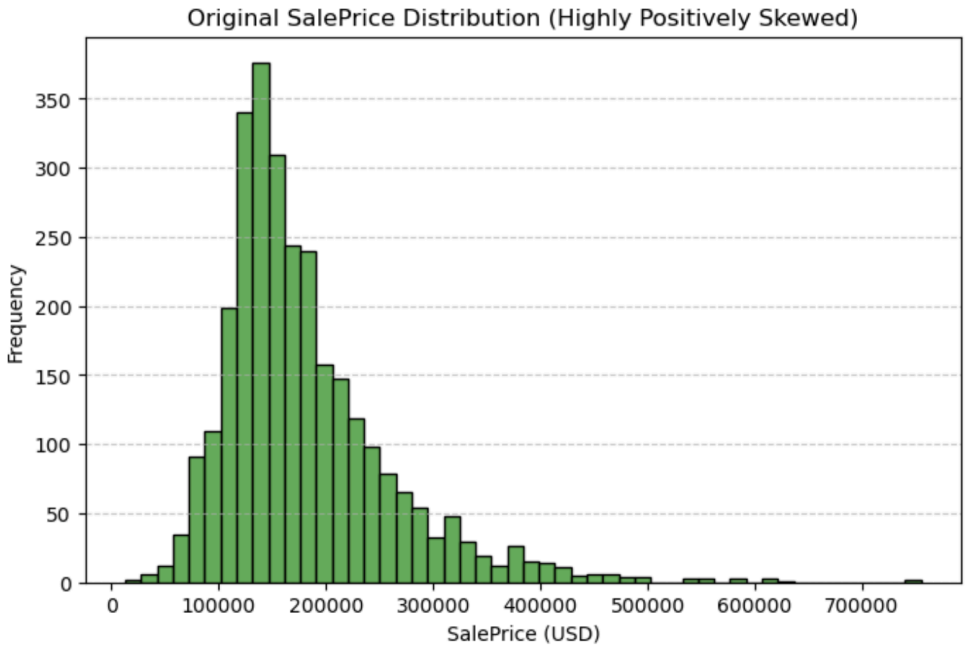
<u>Introducing data</u>

The data used for this learning model is the Ames Housing dataset, which contains 2930 samples and 82 features covering a variety of quantitative and categorical property characteristics. The key features include property size, overall quantity/condition, number of rooms, and key amenities, while our target is sale price.

Visual analysis of data

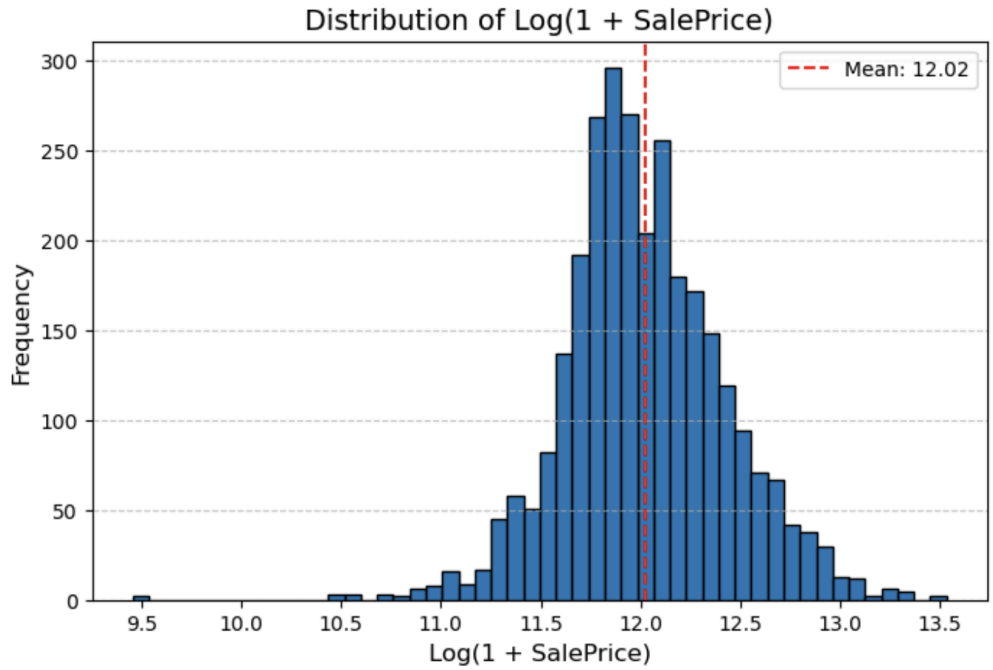
Original target distribution

The initial distribution of the sales price featured showed a high positive skew. Since linear models like OLS and regression perform best when the target variable is normally distributed this Skew data requires transformation.



Log transformed target distribution

To address the skewedness of the sales price was transformed, using the natural logarithm of (1 + SalePrice). The transformation significantly normalized the distribution, making it more suitable for linear regression modeling. The mean of the log transformed price is approximately 12.02.



Data Processing

To prepare the data I had to take a few steps for my regression model. I started out by dropping non-predictive identification columns like order and PID . Next, I handle the missing values and features where “ NA“ which implies absence of a feature were inputted with the string “none“. And features that were missing numerical values were imputed with zero (0). For missing values in lot frontage they were inputted with the median lot frontage of their respective neighborhoods to capture local trends. Then any remaining minor missing values were filled with the mode or median of the column.

I also did a little bit of feature engineering by creating a totalSF meaning total square footage by summing the total Bsmt SF, 1st Flr SF, and 2nd Flr SF. I also converted features that are numerically encoded, into string type so they behave as categories.

Finally for data splitting and scaling. The data was split into a training set, which was 80% and the testing set was 20%. A standard scaler was then applied to standardize the numerical features to have a mean of zero and a standard deviation of one. This scale is crucial for regularized models like Ridge regression.

Methods

Ordinary Least squares

The ordinary least squares model is a fundamental linear regression algorithm. OLS aims to find the set of weights (W) that minimizes the sum of squared errors (SSE) between the predicted values and actual target values.

The weights are calculated directly using the closed-form normal equation where X is the feature matrix and Y is the target vector. OLS serves as an unregularized linear baseline. However, it is highly sensitive to multicollinearity and noise which is expected to be present in a high dimensional dataset.

$$w = (X^T X)^{-1} X^T y$$

<u>Ridge regression</u>

Ridge regression is an extension of OLS that incorporates L2 regularization into loss function to overcome OLS limitations, primarily by addressing multicollinearity and preventing overfitting; the ridge model minimizes the following regularized loss function.

$$Loss = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + \alpha \sum_{j=1}^{p} w_j^2$$

Ridge regression is a regularized form of linear regression that addresses issues of overfitting and multicollinearity by adding a penalty term to the loss function:

$$\alpha \sum_{j=1}^{p} w_j^2$$

This term penalizes large coefficient magnitude. The strength of this penalty is controlled by the hyperparameter alpha, which is the regularization strength, hyperparameter. A larger alpha forces the weight to become smaller, which reduces the model's complexity and makes it less prone to overfitting and more stable when dealing with correlated features; the optimal weights W are found by solving the modified normal equation where the regularization term is added to the X^TX matrix. Ridge regression was an ideal choice for this specific problem because the pre-processing step introduced many highly correlated features making regularization necessary to improve generalization.

Results

The primary objective was to evaluate the predictive accuracy of Ordinary Least Squares, and Ridge regression models trained on the process Ames Housing dataset. This preparation involved log transforming the target variable sales price. The complete data set was divided into 80% training set in 20% testing set. Model performance was assessed by using the root mean squared error, which was calculated on the original USD sale price scale after reversing the log transformation using the formula “exp(y) - 1” to provide a tangible measure of error as well as the R squared score reported on the log transformed data.

Model	R-squared (Log Scale)	RMSE (USD Scale)	Regularization (α)
OLS Regression	−3.12×1023	$39,684.34	N/A
Ridge Regression (Initial)	0.8992	$27,743.15	10
Ridge Regression (Tuned Optimal)	0.8952	$27,714.10	5.9948

Initial Observations

The OLS model performed extremely poorly, it yielded a massive negative R^2 and a very high RMSE of approximately $39,684.34. This outcome demonstrates that the unregularized OLS model overfitted the training data so severely that it failed completely on the unseen test data. This confirms the initial hypothesis that a linear model which struggles with the data says high dimensionality and multicollinearity. In contrast, the initial ridge regression model with alpha equaling 10 immediately achieved a R^2 of .8992 and an RMSE of $27,749.15 which is significantly outperforming OLS.

Ridge hyperparameter tuning

To find the most optimal model, I implemented a tune_ridge_alpha function that tested 100 different values for the alpha hyperparameter. The minimum RMSE was achieved at Alpha value of 5.9948 resulting in a final test RMSE of $27,714.10. This is a slight improvement over the initial alpha equals 10 placeholder values. The visualization confirms that alpha equals 5.9948 minimize the RMSE on the test set demonstrating the successful optimization of the regularization strength.

Result analysis

Judging by the matrix, Ridge regression is the clearly superior model for this house price prediction task within an error margin of approximately $27,714 on the test set. This result is a strong validation of the need for a L2 regularization and high dimensional data sets. The unregularized Ordinary Least Squares model performed terribly indicated by its huge negative R squared value. This OLS failure was a direct

result of the correlated features in relation to the data sets 2930 samples, causing the model to assign overly large unstable weights and overfit massively to the training data.

In contrast, Ridge Regression's success stems from its ability to shrink the magnitude of these unstable coefficients by adding a penalty term to the loss function. This penalty of effectively balancing the bias-variance trade-off, allowing the ridge regression model to generalize well to the unseen test data. The subsequent hyperparameter tuning of alpha confirmed that the level of regularization is critical while the initial alpha of 10 was effective; a slightly lower optimal alpha value of 5.9948 was found to provide the best fit and minimal error.

Comparison to support analysis

The comparison between OLS and Ridge regression yielded very different results that support the primary analysis. The failure of OLS demonstrated by the approximate $12,000 difference in RMSE between the initial performance and the best Ridge performance is primary evidence that multicollinearity and overfitting were severe issues in the high dimensional feature set. If multicollinearity were not significant, OLS would have performed comparably to the ridge regression model with any minimal regularization penalty, but its observed failure proves that ridges L2 regularization was essential for stabilizing the feature weights and managing the complexity features.

In addition, the hyper perimeter tuning curve provides compelling evidence for optimization. This process was crucial because the choice of regularization strength alpha directly impacts the models performance by systematically testing a range of alpha values that optimization successfully isolates a specific alpha of 5.9948. This value quantitatively represents the best balance between minimizing the training error and maximizing the models ability to generalize to unseen test data resulting in the minimum achievable RMSE of $27,714.10 for this model type.

Conclusion

In this project, I successfully implemented and compared OLS and ridge regression for predicting house sale prices using the Ames Housing dataset. The results decisively demonstrate that ridge regression specifically with an optimized alpha of 5.9948 is the optimum choice for this data set. Its L2 regularization proved essential for mitigating the challenges of high dimensionality and multicollinearity resulting in a RMSE of approximately $27,714 on the original USD sale price scale. The failure of the unregularized OLS model highlights the critical role of regularization in building stable and generalizable predictive models when working with complex high dimensional data.

What I learned

I learned the importance of log transformation for highly skewed, target variables as it successfully normalized the sale price distribution, which is necessary to satisfy the assumptions of linear regression to improve model performance. Another important thing I learned was the necessity of using regularization, specifically ridge regression was evident proving that it can effectively combat overfitting and multicollinearity inherent in future rich data sets. The large difference in performance between the baseline OLS model and the regularized model confirmed the practical value of this technique with the OLS models poor performance, serving as a real world example of the pitfalls of a non-regularized approach when dealing with high dimensional complex data.

Challenges

The main challenge was the initial unexpected failure of the OLS model, although ridge regression was planned somewhere as a kind of measure seeing the OLS R squared in the negative range was surprising. This forced a deeper analysis of the feature set to confirm that multicollinearity was the main cause. This was overcome by recognizing that OLS served its purpose as an extreme baseline, validating the need for the regularization of the ridge regression model.

References & Citation

- Devastator, The. *Ames Housing Dataset*. Kaggle, 2024.
- Lee, Minwoo and Hongfei Xue. "Linear Regression". ITCS 3156, University of North Carolina at Charlotte, 2025. Course Lecture.
- Lee, Minwoo and Hongfei Xue. "Gradient Descent and Least Mean Squares". ITCS 3156, University of North Carolina at Charlotte, 2025. Course Lecture.
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Acknowledgment:

I acknowledge referencing code from previous assignments in order to complete this project more specifically “linear regression” and “gradient descent and least means squares” homework.

Source Code: https://github.com/egordo17/IntoToMachineLearingFinalProject